

we assume a constant energy consumption rate  $g$  and a constant charging speed  $\frac{1}{\alpha}$ . The energy consumption for edge  $(i, j)$  is thus given by  $c_{ij} = ga_{ij}$ , the time required to recharge the energy consumed by traveling along edge  $(i, j)$  is given by  $f_{ij} = \alpha c_{ij}$ .

We assume a commercial EV fleet consisting of  $K$  homogeneous EVs, each with a load capacity  $Q$  and a battery capacity  $C$ . The time required to fully recharge the battery from 0 is defined as  $B = \alpha C$ . At the beginning of the planning horizon, all the EVs are at the depot (node 0) with a full battery. All EVs should return to the depot (node  $N + S + 1$ ) before the end of the planning horizon. During each visit to a station or to the depot, an EV can either pay to charge its battery or make profits by injecting energy back to the grid from its battery. The charging cost and discharging reward vary according to the time period (time-of-use energy pricing). We assume that EVs are allowed to perform either charging or discharging during their station visits, but are not allowed to perform both during the same time period.

The planning horizon is formed of  $|T|$  consecutive discrete periods, each of length  $\delta$ . Each time period  $t$  refers to a time interval  $[\delta(t-1), \delta t)$  and is associated with a buying (from the grid) energy price  $p_b^t$  and a selling (to the grid) energy price  $p_s^t$ . We assume that if an EV is to charge or discharge during a time period, then it has to do so for the full time period. Due to the constant linear charging/discharging assumption, a fixed amount of energy  $\frac{\delta}{B}C$  would be charged/discharged during each period, we therefore can calculate a charging cost  $P_{re}^t = \frac{\delta}{B}Cp_b^t$  and a discharging reward  $P_{dis}^t = \frac{\delta}{B}Cp_s^t$  for each period. EVs are recharged back to full battery capacity during the night using at a lower energy price  $p_{night}$ . For the ease of presentation, we define  $P_{night} = \frac{p_{night}}{\alpha}$  as the cost of charging per unit of time during the night. The notations are summarized in Table 2.

The problem formulation that is proposed next jointly optimizes the routes and charging/discharging schedule of the  $K$  vehicles by minimizing the net electricity cost given that all customer demands are satisfied.

### 3.2. Problem Formulation

To formulate EVRPTW-TP, the following binary variables are introduced:

$$\begin{aligned} x_{ijk} &= \begin{cases} 1 & \text{if edge } (i, j) \text{ is traveled by vehicle } k, \\ 0 & \text{otherwise,} \end{cases} \\ r_{itk} &= \begin{cases} 1 & \text{if vehicle } k \text{ charges its battery at node } i \text{ at time period } t, \\ 0 & \text{otherwise,} \end{cases} \\ d_{itk} &= \begin{cases} 1 & \text{if vehicle } k \text{ discharges its battery at node } i \text{ at time period } t, \\ 0 & \text{otherwise.} \end{cases} \end{aligned}$$

The continuous variables are

$\tau_{ik}$ : arrival time of vehicle  $k$  at node  $i$ ,

**Table 2** Summary of the Notation

|              | Definition                                                                    |
|--------------|-------------------------------------------------------------------------------|
| $V_c$        | Set of customer nodes $V_c = \{1, 2, \dots, N\}$                              |
| $V_s$        | Set of station nodes $V_s = \{N + 1, \dots, N + S\}$                          |
| $V_{od}$     | Set of depot nodes $V_{od} = \{0, N + S + 1\}$                                |
| $V_{c,s}$    | $V_{c,s} = V_c \cup V_s$                                                      |
| $V_{c,s,o}$  | $V_{c,s,o} = V_c \cup V_s \cup \{0\}$                                         |
| $V_{c,s,d}$  | $V_{c,s,d} = V_c \cup V_s \cup \{N + S + 1\}$                                 |
| $V_{s,d}$    | $V_{s,d} = V_s \cup \{N + S + 1\}$                                            |
| $V_{s,o}$    | $V_{s,o} = V_s \cup \{0\}$                                                    |
| $V_{s,od}$   | $V_{s,od} = V_s \cup V_{od}$                                                  |
| $V_{c,od}$   | $V_{c,od} = V_c \cup V_{od}$                                                  |
| $V_{c,s,od}$ | $V_{c,s,od} = V_c \cup V_s \cup V_{od}$                                       |
| $p$          | Depot node $p = N + S + 1$                                                    |
| $E$          | Set of all edges                                                              |
| $T$          | Set of charging/discharging periods                                           |
| $N$          | Number of customers                                                           |
| $S$          | Number of stations                                                            |
| $K$          | Number of EVs                                                                 |
| $\delta$     | Length of each charging/discharging period                                    |
| $Q$          | Cargo capacity                                                                |
| $C$          | Battery capacity                                                              |
| $B$          | Amount of time required to fully charge the EV battery from empty             |
| $\alpha$     | The reciprocal of the constant charging speed                                 |
| $g$          | Energy consumption rate with respect to distance traveled                     |
| $a_{ij}$     | Travel distance of edge $(i, j)$                                              |
| $t_{ij}$     | Travel time of edge $(i, j)$                                                  |
| $c_{ij}$     | Energy consumption along edge $(i, j)$                                        |
| $f_{ij}$     | The amount of time required to charge the energy consumed along edge $(i, j)$ |
| $e_i$        | Earliest service start time at node $i$                                       |
| $l_i$        | Latest service start time at node $i$                                         |
| $s_i$        | Required service time at node $i$                                             |
| $q_i$        | The demand at node $i$                                                        |
| $p_{night}$  | Unit electricity buying (from the grid) price at night (¢/kWh)                |
| $p_b^t$      | Unit electricity buying (from the grid) price during period $t$ (¢/kWh)       |
| $p_s^t$      | Unit electricity selling (to the grid) price during period $t$ (¢/kWh)        |
| $P_{night}$  | The cost of charging per unit of time at night                                |
| $P_{re}^t$   | Cost of charging during period $t$                                            |
| $P_{dis}^t$  | Reward for discharging during period $t$                                      |

$b_{ik}$  : remaining energy (in terms of charging time) in vehicle  $k$  upon arrival to node  $i$ ,

$u_{ik}$  : remaining cargo in vehicle  $k$  upon arrival to node  $i$ .

The decision variables  $r_{itk}$  and  $d_{itk}$  are only associated with the station and depot nodes, while the remaining variables are associated with all the nodes. EVRPTW-TP is formulated as

$$\min \sum_{k=1}^K \sum_{i \in V_{s,od}} \sum_{t \in T} [r_{itk} P_{re}^t - d_{itk} P_{dis}^t] + \sum_{k=1}^K P_{night} [B - b_{pk} - \sum_{t \in T} \delta(r_{ptk} - d_{ptk})], \quad (1)$$

$$\text{s.t.} \quad \sum_{k=1}^K \sum_{j \in V_{c,s,d}} x_{ijk} = 1, \quad \forall i \in V_c, \quad (2)$$

$$\sum_{i \in V_{c,s,o}} x_{ipk} = 1, \quad \forall k \in \{1, 2, \dots, K\}, \quad (3)$$

$$\sum_{j \in V_{c,s,o}} x_{jik} - \sum_{j \in V_{c,s,d}} x_{ijk} = 0, \quad \forall i \in V_{c,s}, k \in \{1, 2, \dots, K\}, \quad (4)$$

$$\tau_{ik} + (t_{ij} + s_i)x_{ijk} - \delta|T|(1 - x_{ijk}) \leq \tau_{jk}, \quad \forall i \in V_{c,s,o}, j \in V_{c,s,d}, k \in \{1, 2, \dots, K\}, \quad (5)$$

$$t\delta(r_{itk} + d_{itk}) + t_{ij}x_{ijk} - \delta|T|(1 - x_{ijk}) \leq \tau_{jk}, \quad \forall i \in V_{s,o}, j \in V_{c,s,d}, t \in T, k \in \{1, 2, \dots, K\}, \quad (6)$$

$$e_i \leq \tau_{ik} \leq l_i, \quad \forall i \in V_{c,s,od}, k \in \{1, 2, \dots, K\}, \quad (7)$$

$$b_{0k} = B, \quad \forall k \in \{1, 2, \dots, K\}, \quad (8)$$

$$b_{jk} \leq b_{ik} - f_{ij}x_{ijk} + B(1 - x_{ijk}), \quad \forall i \in V_c, j \in V_{c,s,d}, k \in \{1, 2, \dots, K\}, \quad (9)$$

$$b_{jk} \leq b_{ik} + \sum_{t \in T} \delta r_{itk} - \sum_{t \in T} \delta d_{itk} - f_{ij}x_{ijk} + B(1 - x_{ijk}), \quad \forall i \in V_{s,o}, j \in V_{c,s,d}, k \in \{1, 2, \dots, K\}, \quad (10)$$

$$\sum_{t \in T} \delta r_{itk} \leq B - b_{ik}, \quad \forall i \in V_{s,od}, k \in \{1, 2, \dots, K\}, \quad (11)$$

$$\sum_{t \in T} \delta d_{itk} \leq b_{ik}, \quad \forall i \in V_{s,od}, k \in \{1, 2, \dots, K\}, \quad (12)$$

$$r_{itk} + d_{itk} \leq 1, \quad \forall i \in V_{s,od}, t \in T, k \in \{1, 2, \dots, K\}, \quad (13)$$

$$\tau_{ik} - (t-1)\delta \leq \delta|T|(1 - d_{itk} - r_{itk}), \quad \forall i \in V_{s,d}, t \in T, k \in \{1, 2, \dots, K\}, \quad (14)$$

$$u_{jk} \leq u_{ik} - q_i x_{ijk} + Q(1 - x_{ijk}), \quad \forall i \in V_{c,s,o}, j \in V_{c,s,d}, k \in \{1, 2, \dots, K\}, \quad (15)$$

$$u_{0k} = Q, \quad k \in \{1, 2, \dots, K\}, \quad (16)$$

$$0 \leq b_{jk} \leq B \sum_{i \in V_{c,s,o}} x_{ijk}, \quad \forall j \in V_s, k \in \{1, 2, \dots, K\}, \quad (17)$$

$$u_{ik}, \tau_{ik} \geq 0, \quad \forall i \in V_{c,s,od}, k \in \{1, 2, \dots, K\}, \quad (18)$$

$$x_{ijk} \in \{0, 1\}, \quad \forall i \in V_{c,s,o}, j \in V_{c,s,d}, k \in \{1, 2, \dots, K\}, \quad (19)$$

$$r_{itk}, d_{itk} \in \{0, 1\}, \quad \forall i \in V_{c,od}, t \in T, k \in \{1, 2, \dots, K\}. \quad (20)$$

The objective function (1) minimizes the net cost, i.e. the total cost of charging the batteries minus the total reward earned from discharging the batteries. The first part of the objective function corresponds to the net cost during the planning horizon, while the second part is the cost of fully recharging back all EVs at night. Constraints (2) ensure that every customer is served by exactly one EV. Constraints (3) force all the EVs to return to the depot by the end of the

planning horizon. Constraints (4) guarantee that no route ends at a customer or a station node. Constraints (5) ensure time feasibility of the edges leaving the customer and the depot nodes, while constraints (6) deal with the edges originating from charging stations. Constraints (7) ensure that the time windows of all the nodes are not violated. Constraints (8) indicate that every EV is fully charged before leaving the depot. Constraints (9)–(10) track the battery capacity along the route. Constraints (11) indicate that an EV battery cannot be recharged to a level that exceeds its capacity, while Constraints (12) state that an EV battery cannot be discharged to a level below 0. Constraints (13) indicate that an EV is allowed to discharge or recharge its battery at a station or the depot nodes but is not allowed to discharge and recharge during the same time period. Constraints (14) ensure that an EV cannot start charging/discharging at a station/depot before arrival and before the start of the time period. Constraints (15) ensure that the demands along a route are all satisfied, whereas Constraints (16) state that all EVs have a full cargo at the start of the planning horizon. Constraints (17)–(20) indicate the variables types and limits.

#### 4. Lagrangian Relaxation for the EVRPTW-TP

As shown in Section 6.3, solving EVRPTW-TP to optimality is computationally very challenging. Lagrangian relaxation is a well known algorithm that has been used to address many complex optimization problems. Particularly, in the context of vehicle routing, Lagrangian relaxation has been used to address several variants of VRP (Fisher et al. 1997, Kallehauge et al. 2006).

For the EVRPTW-TP, all the constraints other than Constraints (2) are associated with a particular vehicle  $k$ . Given this special structure which is common in vehicle routing problems, Constraints (2) are relaxed and the violation is penalized in the objective function using the Lagrangian multipliers  $\lambda_i$ . The resulting relaxed problem is

$$\begin{aligned}
 Z_{LR}(\lambda) = \min \quad & \sum_{k=1}^K \sum_{i \in V_{s,od}} \sum_{t \in T} [r_{itk} P_{re}^t - d_{itk} P_{dis}^t] \\
 & + \sum_{k=1}^K P_{night} [B - b_{pk} - \sum_{t \in T} \delta(r_{ptk} - d_{ptk})] \\
 & + \sum_{i \in V_c} \lambda_i (1 - \sum_{k=1}^K \sum_{j \in V_{c,s,d}} x_{ijk}), \\
 \text{s.t.} \quad & (3) - (20).
 \end{aligned}$$

Since the EV fleet is homogeneous, problem  $Z_{LR}(\lambda)$  decomposes into  $K$  identical sub-problems

$$\begin{aligned}
 Z_{SP}(\lambda) = \min \quad & \sum_{i \in V_{s,od}} \sum_{t \in T} [r_{it} P_{re}^t - d_{it} P_{dis}^t] + P_{night} [B - b_p - \sum_{t \in T} \delta(r_{pt} - d_{pt})] \\
 & - \sum_{i \in V_c} \lambda_i \sum_{j \in V_{c,s,d}} x_{ij}, \\
 \text{s.t.} \quad & (3) - (20).
 \end{aligned}$$